

HW01

ENAE 601: Astrodynamics

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Problem 1**2.7**

Starting with Eq. 2.35a (shown below), prove that $\dot{\mathbf{r}} = \mathbf{v} \cdot \hat{\mathbf{u}}_r$ and interpret this result.

$$\mathbf{r} \cdot \dot{\mathbf{r}} = r\dot{r}$$

answer

Problem 2**2.8**

Show that $\hat{\mathbf{u}}_r \cdot \frac{d\hat{\mathbf{u}}_r}{dt} = 0$, where $\hat{\mathbf{u}}_r = \frac{\mathbf{r}}{r}$. Use only the fact that $\hat{\mathbf{u}}_r$ is a unit vector. Interpret this result.

answer

Problem 3**2.9**

problem

answer

Problem 4**2.10**

problem

answer

Problem 5**2.11**

problem

answer

Problem 6**2.12**

problem

answer

Problem 7**2.13**

problem

answer

Problem 8**2.15**

problem

answer

Problem 9**2.17**

problem

answer

Problem 10**2.22**

problem

answer